

## Module 4

# The Time Value of Money

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FINE 3010, Financial Management  
Prof. Renping Li, Tulane University

# Goals

- **Part 1**
  - Translate dollars today into dollars tomorrow (“future value”)
  - Translate dollars tomorrow into dollars today (“present value”)

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- **Part 3**
  - Special cases: perpetuities, annuities, and their variants

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  - Financial calculators
- **Part 3**
  - Special cases: perpetuities, annuities, and their variants
- **Part 4**
  - The role of inflation

## Module 4, Part 1

# Future Value and Present Value

# The Time Value of Money



**Would you rather receive a dollar today, or a dollar tomorrow?**

# The Time Value of Money

- A dollar today is worth more than a dollar tomorrow



# The Time Value of Money

- A dollar today is worth more than a dollar tomorrow
- Why?



# The Time Value of Money

- A dollar today is worth more than a dollar tomorrow
- Why?
- Because we can invest that dollar today and collect interest on our investment!



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- Why?
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  - Assuming, there is some positive rate of return investment for us to pursue

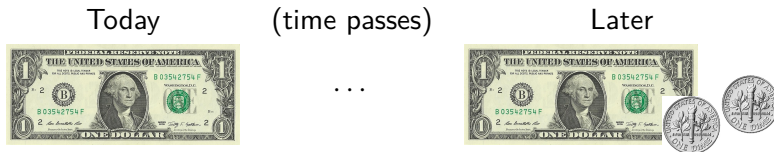


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$$\text{Future value (FV)} = \$100 \times (1 + 0.06)^5 = \$133.82$$

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- Compound interest is the default in finance (e.g., deposit rate)

# Future Value, Compound Interest

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| Year | Start balance | Compound interest             | End balance |
|------|---------------|-------------------------------|-------------|
| 1    | \$100.00      | $0.06 \times 100.00 = \$6.00$ | \$106.00    |
| 2    | \$106.00      | $0.06 \times 106.00 = \$6.36$ | \$112.36    |
| 3    | \$112.36      | $0.06 \times 112.36 = \$6.74$ | \$119.10    |
| 4    | \$119.10      | $0.06 \times 119.10 = \$7.15$ | \$126.25    |
| 5    | \$126.25      | $0.06 \times 126.25 = \$7.57$ | \$133.82    |

## Future Value, Compound Interest

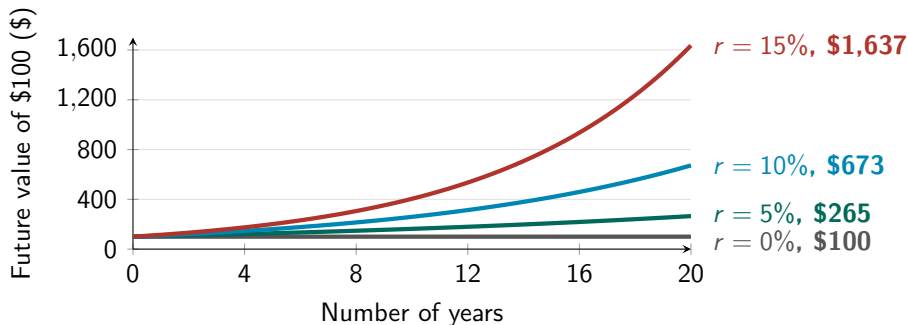
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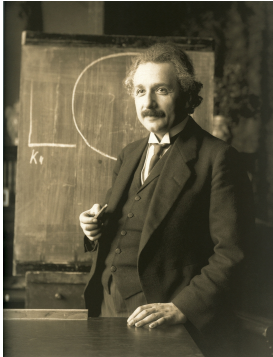
- With compound interest, your \$100 grows to  $\$100(1 + r)^T = 133.82$ , so you earned \$33.82 in compound interest

# Future Value

- The future value (FV) depends heavily on the interest rate  $r$ , especially as the time dimension  $T$  grows

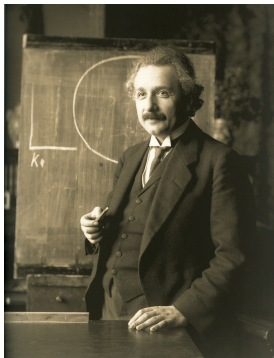


# What's the Most Powerful Force in the Universe?



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$$\text{Return} = \frac{\$180 - \$100}{\$100} = 80\%$$

# Holding Period Return

- **Holding period return**, percentage growth over the period an asset is owned, which can be longer or shorter than a year, e.g., owning a stock for  $T$  periods with returns  $r_1, r_2, \dots, r_T$  gives  $1 + r_{\text{holding period}} = (1 + r_1) \times (1 + r_2) \times \dots (1 + r_T)$

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$$r_{\text{holding period}} = (1 + r_1) \times (1 + r_2) - 1 = 1.80 \times 0.80 - 1 = 44\%$$

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- The annualized return is *not*  $\frac{80\% - 20\%}{2} = 30\%$

## Recap, Three Kinds of Returns

| Return                                                         | Holding period return                                                                    | Annualized return                                                                     |
|----------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Percentage growth of an asset's value over some period of time | Percentage growth over the whole period an asset is owned, longer or shorter than a year | Percentage growth per year, a geometric mean over the $T$ years of the holding period |

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- These financial returns are *not directly related in any way* with **Module 3's** accounting-based ROA and ROE measures

## Example, Future Value

- Peter Minuit bought Manhattan Island for \$24 in 1626
- a. What would that investment have been worth in 2021, if it had earned an 8% annual return?
- b. What if the return had been 3.5%?



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b.  $FV = \$24 \times (1 + 0.035)^{395} = \$19 \text{ million}$

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  - We will learn how to calculate this with the financial calculator



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- \$100 in the future is worth less than \$100 in the present, so we “discount” the value of the future cash flow to its present value

# Present Value

- **Discount rate**, the rate used to compute PVs (i.e.,  $r$ )
  - For a guaranteed future cash flow, the correct discount rate is the risk-free interest rate
  - For a risky future cash flow, we may need a higher discount rate to account for the higher risk, we will return to this in **Modules 8 and 9**

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- **Discount factor**, the present value of \$1 received in year  $T$  when the discount rate is  $r$ , less than one when  $r$  is positive

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 **Discount factor**

- **Discounted cash flow (DCF)**, the method of calculating present values by discounting future cash flows

## Example, Present Value

- Suppose you need \$3,000 next year to buy a new computer. The interest rate is 8% per year. How much money should you set aside now in order to pay for the purchase?

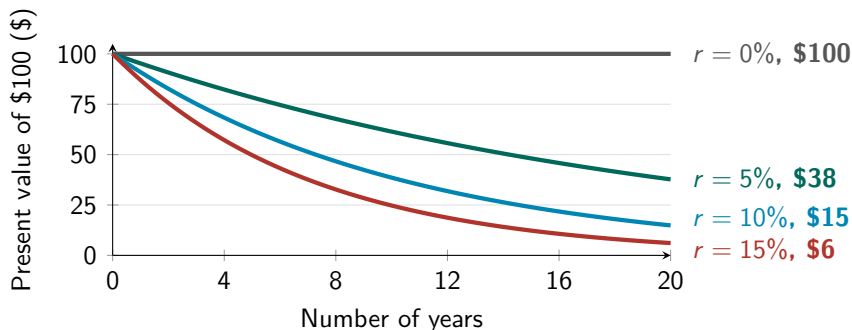
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$$\text{Present value (PV)} = \frac{\text{FV}}{(1 + r)^t} = \frac{\$3,000}{(1 + 0.08)^1} = \$2,778$$

# Present Value

- The present value (PV) of \$100 in the future depends heavily on the discount rate  $r$ , especially as the time dimension  $T$  grows



# Present Value, Multiple Future Cash Flows

- Present values can be added together to calculate the total PV of multiple future cash flows

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$$PV_{\text{total}} = \frac{C_1}{(1+r)^1} + \frac{C_2}{(1+r)^2} + \dots$$

- where  $C_1$  is the cash flow in one year,  $C_2$  is the cash flow in two years, etc.

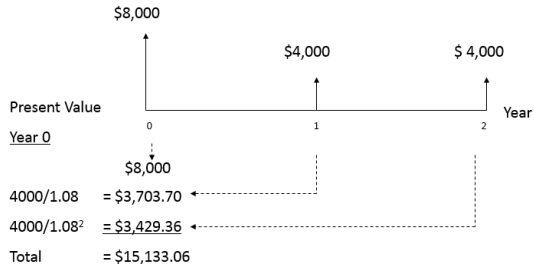
## Example, Multiple Future Cash Flows

- Your auto dealer gives you the choice to either
  - A. Pay \$15,500 cash now, or
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- If the discount rate is 8%, which do you prefer?

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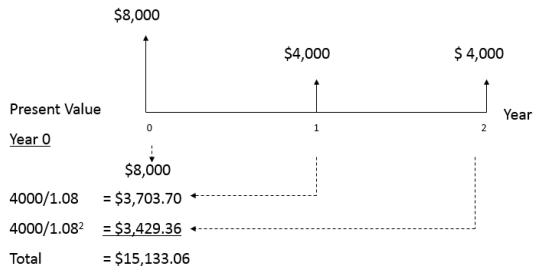
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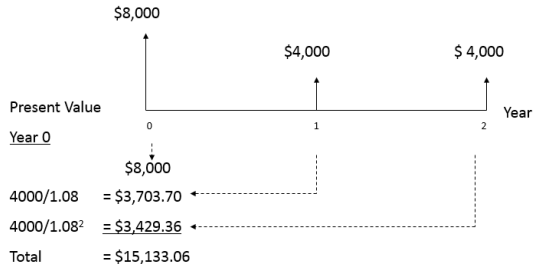
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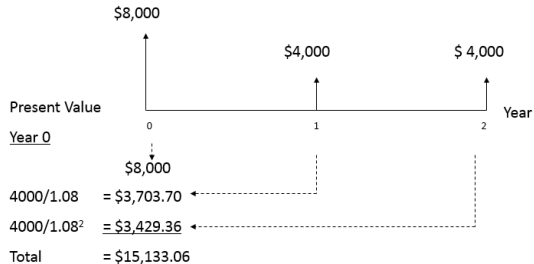


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- The present value (\$15,133.06) is less than \$15,500, so choose the payment plan
- This is the principle behind the NPV rule in capital budgeting, which we will return to in **Modules 6 and 7**

# Recap of Part 1

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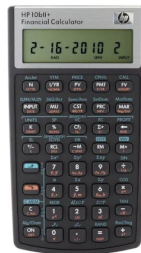
- Present values can be added together to calculate the total present value of multiple future cash flows

## Module 4, Part 2

# Financial Calculators

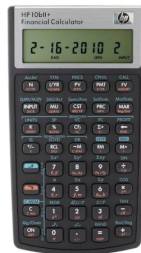
# Financial Calculators

- Speeds up time value of money problems (and solves harder ones!)



# Financial Calculators

- Speeds up time value of money problems (and solves harder ones!)
- A complement to the underlying math and intuition, never a substitute



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- After you reset, you should repeat the next two settings!

# Set the Number of Significant Digits to at Least 4!

- On the HP 10bII+, to change the number of digits displayed to 4 decimals, press
  - OrangeOption
  - = (DISP)
  - 4

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  - 2nd
  - .
  - 4
  - ENTER

# Set the Number of Compounding Periods per Year to 1!

- On the HP 10bII+, to set it to 1,  
press
  - 1
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  - PMT (P/YR)

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- On the HP 10bII+, to set it to 1, press
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- To change back to the factory default (12), press
  - 12
  - OrangeOption
  - PMT (P/YR)

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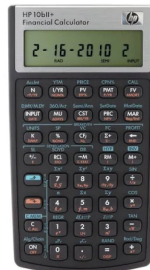
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  - OrangeOption
  - PMT (P/YR)
- To change back to the factory default (12), press
  - 12
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- On the TI BA II Plus, press
  - 2nd
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  - ENTER

# Set the Number of Compounding Periods per Year to 1!

- On the HP 10bII+, to set it to 1, press
  - 1
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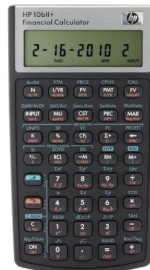
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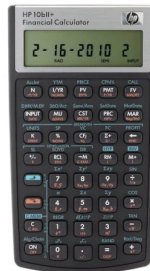
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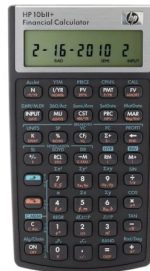
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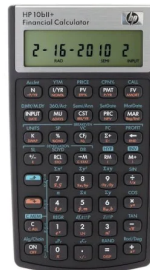
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  - 0 PMT
  - 3000 FV
  - PV

|   |   |    |     |      |
|---|---|----|-----|------|
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# The Equation the Calculator Solves

- We solved this problem earlier in this module using the following formula

$$\text{Present value} = PV = \frac{FV}{(1 + r)^t} = \frac{\$3,000}{(1 + 0.08)^1} = \$2,778$$

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- In 2007, Puerto Rico needed to borrow \$2.6 billion for 47 years
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- These bonds were sold at \$94.40 each (they sold around  $\$2.6 \text{ billion} / 94.40 = 27.5 \text{ million}$  bonds, and thus promised to pay back around \$27.5 billion in total)
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- Solution,
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- Bonus, what would the payment be if the monthly interest was 0.2%?

## Example, Pick Your Payment!

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- Suppose Joe and Lucy both buy cars for \$20,000, and their monthly interest is 0.6% (just over 7% per year). Joe picks a payment of \$400, Lucy picks a payment of \$600. How many months will Joe and Lucy be making payments on their cars?

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- Let's try to solve some of the previous examples using Excel!

## Example, A-Rod's \$275 Million Contract with the Yankees

| Year         | Salary     | "Signing bonus" |
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| 2008         | 27         | 2               |
| 2009         | 32         | 1               |
| 2010         | 32         | 1               |
| 2011         | 31         | 1               |
| 2012         | 29         | 1               |
| 2013         | 28         | 1               |
| 2014         | 25         | 3               |
| 2015         | 21         |                 |
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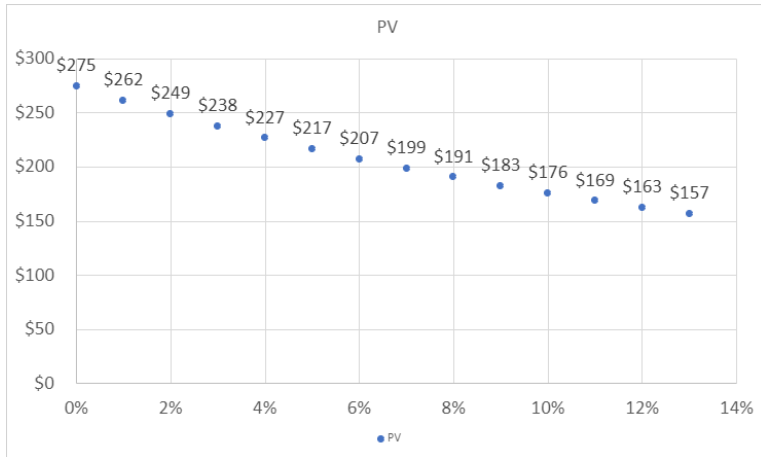
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- What was A-Rod's \$275 million contract with the Yankees actually worth?
- Suppose the discount rate is 10%. What is the present value of his contract?
- Why might it be useful for A-Rod to know the PV, and not simply the (undiscounted) sum of all future payments?



# The PV of Payments to A-Rod



## Recap of Part 2

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- Set the number of compounding periods per year to 1, and show at least 4 significant digits

## Recap of Part 2

- Financial calculators solve any one of N, I/YR, PV, PMT, and FV from the other four
- The PV has the opposite sign of the payments (PMT) and the future value (FV)
- Set the number of compounding periods per year to 1, and show at least 4 significant digits
- Excel has the same functionality, with the rate in decimals

## Module 4, Part 3

# Special Cases: Perpetuities and Annuities

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- Why care about perpetuities and annuities?
  - Many real-world cash flows have these patterns (particularly annuities, e.g., most mortgages or other fixed-term loans)
  - There exist neat shortcuts that make it very easy to calculate the present value of these patterns of cash flows

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- A perpetuity is a stream of equal cash flows ( $C$ ) that occur at regular intervals and last forever



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- $C$  is the annual cash flow (aka the payment), and  $r$  is the discount rate

## Example, Perpetuities

- Recall A-Rod's Yankees contract. Suppose the discount rate is 10%, then the PV of the contract (paying \$275m over 10 years) is \$175.6m
- Now suppose the Yankees offer A-Rod two options,
  - A. The contract as previously described (10 years with \$275m in total), or
  - B. A perpetual payment of  $C$  dollars every year. He will play for 10 years, but continue to be paid forever, as long as he lives, and then his children, and then his children's children, etc.
- How much should the Yankees offer as the perpetual payment  $C$  for him to be **indifferent** between contract (A) and the “being paid forever” contract (B)?



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$$PV_{\text{Contract A}} = PV_{\text{Contract B}} = \frac{C_B}{r} \Rightarrow C_B = PV \times r = 175.6 \times 10\% = \$17.56m$$

# Perpetuity, Timing

- **Perpetuity due**, first payment at time 0

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- **Ordinary perpetuity**, first payment at time 1

$$PV = \frac{C}{r}$$



# Perpetuity, Timing

- **Perpetuity due**, first payment at time 0

$$PV = \frac{C}{r} \times (1 + r)$$



- **Ordinary perpetuity**, first payment at time 1

$$PV = \frac{C}{r}$$



- **Delayed perpetuity**, first payment at time  $t > 1$

$$PV = \frac{C}{r} \times \frac{1}{(1+r)^{t-1}}$$



## Growing Perpetuities

- Suppose you have a perpetual cash flow that starts at  $C$  but grows at a constant growth rate  $g$  forever. This is a growing perpetuity

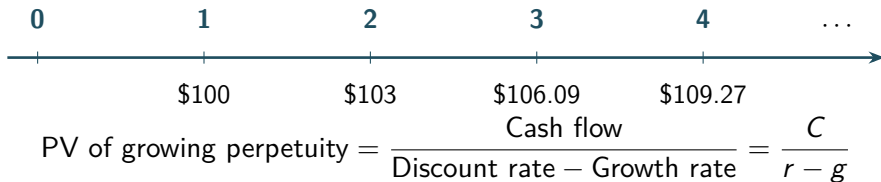
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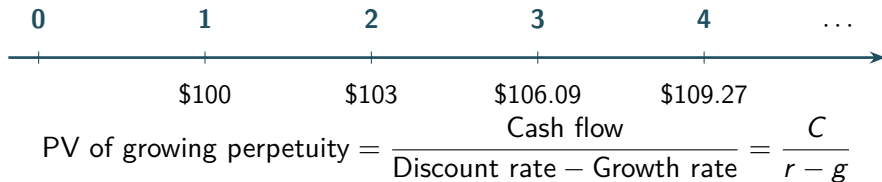
## Growing Perpetuities

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- This formula comes up often when valuing stocks, e.g., in the “Gordon growth model” and DCF valuation (more in [Modules 5 and 9](#))
- When  $g = 0$ , the PV simplifies to the formula for a regular (non-growing) perpetuity

# Annuities

- An annuity is a stream of  $N$  equal cash flows paid at regular intervals



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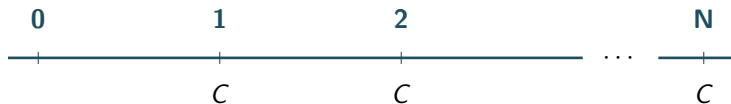


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- $C$  is the annual cash flow (aka the payment),  $r$  is the discount rate, and  $N$  is the number of years that the annuity lasts

## Example, Annuity

- You are purchasing a car. You are scheduled to make 3 annual installments of \$8,000 per year, with the first payment a year from now. Given a discount rate of 10%, what is the price you are paying for the car (i.e., what is the PV of the annuity)?

## Example, Annuity

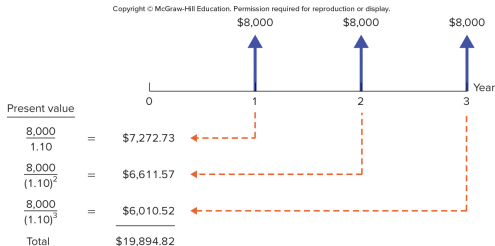
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$$PV = C \frac{1}{r} \left( 1 - \frac{1}{(1+r)^N} \right) = \$8,000 \times \frac{1}{0.10} \times \left( 1 - \frac{1}{(1+0.10)^3} \right) = \$19,895$$

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## Example, Annuity

- Suppose the discount rate for A-Rod's contract is 10%, so the present value is \$175.6m
- Now suppose instead the Yankees offer an annuity for 10 years with the same present value
- How much would he get paid per year, and in total (undiscounted) over 10 years, under such a same-value annuity contract?



## Example, Annuity

- Solution,

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$$175.6 = C \frac{1}{10\%} \left( 1 - \frac{1}{(1 + 10\%)^{10}} \right) \Rightarrow C = 28.58\text{m}$$

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$$175.6 = C \frac{1}{10\%} \left( 1 - \frac{1}{(1 + 10\%)^{10}} \right) \Rightarrow C = 28.58\text{m}$$

- So he would be paid \$28.58m per year, or \$285.8m in undiscounted total over 10 years (slightly more in total, because his actual contract is front-loaded, which raises its present value compared to a constant payment)

## Discuss, Front-Loaded or Level Cash Flows?

- If we have two cash flows with the same undiscounted total, but one is front-loaded, another is equally distributed over the years (i.e., an annuity), which one has a higher present value?

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- If we have two cash flows with the same undiscounted total, but one is front-loaded, another is equally distributed over the years (i.e., an annuity), which one has a higher present value?
- Answer, the front-loaded one
- What about a back-loaded cash flow and an annuity?

# Perpetuities and Annuities, Timing

- **Annuity due**, first payment at time 0

$$PV = PV \text{ annuity} \times (1 + r)$$



# Perpetuities and Annuities, Timing

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- **Annuity due**, first payment at time 0  $PV = PV \text{ annuity} \times (1 + r)$



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- **Delayed annuity**, first payment at time  $t > 1$   $PV = PV \text{ annuity} \times \frac{1}{(1+r)^{t-1}}$



## Practice Problem

- A store offers two payment plans. Under the installment plan, you pay 25% down and 25% of the purchase price in each of the next 3 years. If you pay the entire bill immediately, you can get a discount of 7% on the purchase price. Assume the product sells for \$100
- a-1. Calculate the present value of the payments if you can borrow or lend funds at an interest rate of 4%
- a-2. Which is a better deal?
- b-1. Calculate the present value if the payments on the 4-year installment plan do not start for a full year
- b-2. Which is a better deal?

## Practice Problem

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- a-2. Paying in full now is better (\$93 beats \$94.38)
- On a financial calculator,  $PMT = -25$ ,  $N = 3$ ,  $FV = 0$ ,  $I/YR = 4$ , leading to 69.38, and next,  $69.38 + 25 = 94.38$

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- On a financial calculator,  $PMT = -25$ ,  $N = 4$ ,  $FV = 0$ ,  $I/YR = 4$

## Practice Problem

- A local bank advertises the following deal, “Pay us \$100 at the end of each year for 11 years, and then we will pay you (or your beneficiaries) \$100 at the end of each year forever”
  - a. Calculate the present value of your payments to the bank if the interest rate is 7.00%
  - b. What is the present value of a \$100 perpetuity deferred for 11 years if the interest rate is 7.00%?
  - c. Is this a good deal?

## Practice Problem

- a. Your payments to the bank are an 11-year ordinary annuity

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$$PV = C \left( \frac{1}{r} \right) \left( 1 - \frac{1}{(1+r)^{11}} \right)$$
$$PV = 100 \left( \frac{1}{7\%} \right) \left( 1 - \frac{1}{(1+7\%)^{11}} \right) = \$749.83$$

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$$PV_{11} = \frac{C}{r}$$
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- c. Not a good deal (you pay \$749.83 for something worth \$678.70)

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$$\text{PV of growing annuity} = \frac{C}{r - g} \left[ 1 - \left( \frac{1 + g}{1 + r} \right)^N \right]$$

# Take the Lump Sum or Growing Annuity?

- The Robinsons of Munford, Tennessee were one of the three winners in the “1.6 billion” Powerball jackpot in 2016
- They could choose between,
  - A. A lump sum of \$327 million, or
  - B. 30 installments of a growing annuity due with  $g = 4\%$ , starting at \$9.43m, with 30 payments, totaling around \$529 million



Photo, AP/Mark Humphrey

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- Which option should they take?



Photo, AP/Mark Humphrey

# Take the Lump Sum or Growing Annuity?

- Not evenly divided, but a growing annuity due ( $g = 4\%$ ), it starts immediately and grows for another 29 years at 4% and then ends

| Year         | Payout        |
|--------------|---------------|
| 0            | 9.43          |
| 1            | 9.81          |
| 2            | 10.20         |
| ...          | ...           |
| 27           | 27.19         |
| 28           | 28.28         |
| 29           | 29.41         |
| <b>Total</b> | <b>528.88</b> |

# Take the Lump Sum or Growing Annuity?

- Not evenly divided, but a growing annuity due ( $g = 4\%$ ), it starts immediately and grows for another 29 years at 4% and then ends
- Suppose the discount rate is 3%

| Year         | Payout        |
|--------------|---------------|
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| 1            | 9.81          |
| 2            | 10.20         |
| ...          | ...           |
| 27           | 27.19         |
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# Take the Lump Sum or Growing Annuity?

- The PV of the growing annuity due is

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$$= \frac{9.43}{0.03 - 0.04} \left[ 1 - \left( \frac{1 + 0.04}{1 + 0.03} \right)^{30} \right] \times (1 + 0.03)$$

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$$\begin{aligned} & \frac{C}{r-g} \left[ 1 - \left( \frac{1+g}{1+r} \right)^N \right] \times (1+r) \\ &= \frac{9.43}{0.03-0.04} \left[ 1 - \left( \frac{1+0.04}{1+0.03} \right)^{30} \right] \times (1+0.03) \\ &= \$326.58\text{m} \end{aligned}$$

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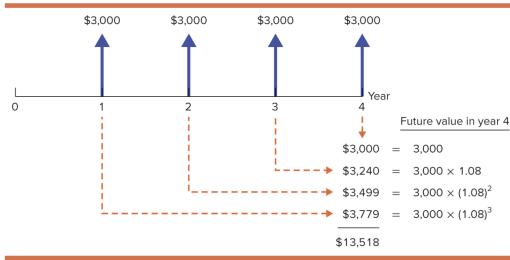
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- That is, the present value of the annuity is almost exactly the same as (but slightly lower than) the \$327m lump sum

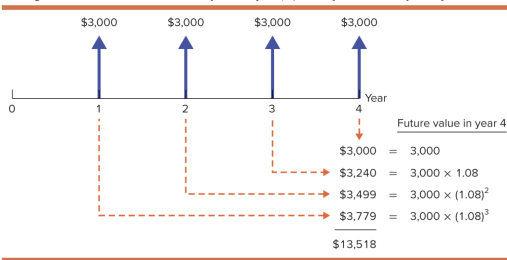
# Future Value of Annuities

FIGURE 5.11 Calculating the future value of an ordinary annuity of \$3,000 a year for four years (interest rate = 8%).



# Future Value of Annuities

FIGURE 5.11 Calculating the future value of an ordinary annuity of \$3,000 a year for four years (interest rate = 8%).



$$\text{FV of annuity of \$1 a year} = \text{PV of annuity of \$1 a year} \times (1 + r)^N$$

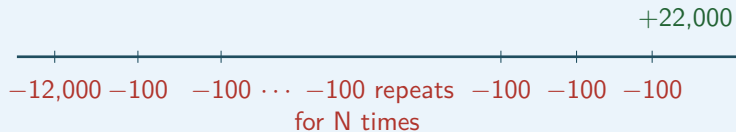
$$= \left[ \frac{1}{r} - \frac{1}{r(1 + r)^N} \right] \times (1 + r)^N = \frac{(1 + r)^N - 1}{r}$$

## Practice Problem

- I now have \$12,000 in the bank earning interest of 0.50% per month. I need \$22,000 to make a down payment on a house. I can save an additional \$100 per month. How long will it take me to accumulate the \$22,000?

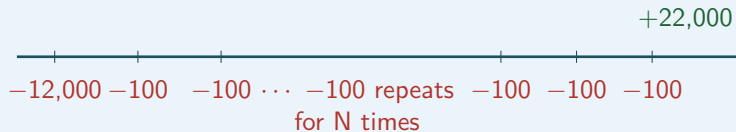
## Practice Problem

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- You need to solve, future value (FV) of the \$12,000 + FV of the N \$100s = FV of the \$22,000

$$12000 \cdot (1 + 0.005)^N + 100 \cdot \frac{(1 + 0.005)^N - 1}{0.005} = 22000$$

## Practice Problem

- The problem can, alternatively, be solved as, present value (PV) of the \$12,000 + PV of the N \$100s = PV of the \$22,000

|           |          |            |           |            |           |
|-----------|----------|------------|-----------|------------|-----------|
| Enter     |          | 0.50       | -12,000   | -100       | 22,000    |
|           | <b>N</b> | <b>I/Y</b> | <b>PV</b> | <b>PMT</b> | <b>FV</b> |
| Solve for | 54.52    |            |           |            |           |

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- Let's recall that financial calculators are programmed to solve the following,

$$PV + \left[ \frac{PMT}{(1 + I)^1} + \frac{PMT}{(1 + I)^2} \cdots + \frac{PMT}{(1 + I)^N} \right] + \frac{FV}{(1 + I)^N} = 0$$

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| Solve for | 54.52    |            |           |            |           |

## Practice Problem

- A couple will retire in 50 years. They plan to spend about \$20,000 a year in retirement, which should last about 25 years. They believe that they can earn a real interest rate of 9% on retirement savings
  - a. If they make annual payments into a savings plan, how much will they need to save each year? Assume the first payment comes in 1 year
  - b. How would the answer to part (a) change if the couple also realize that in 20 years they will need to spend \$50,000 on their child's college education?

# Practice Problem

- a. The cash flow is

Time 0, today

Time of retirement

Final withdrawal



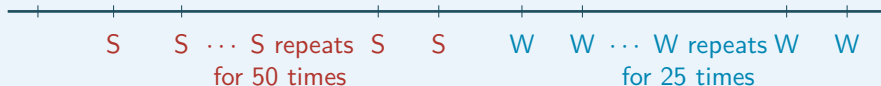
## Practice Problem

- a. The cash flow is

Time 0, today

## Time of retirement

## Final withdrawal



- How much savings is required at the time of retirement?

$$PV_{\text{withdrawals, year 50}} = 20,000 \frac{1}{0.09} \left( 1 - \frac{1}{(1 + 0.09)^{25}} \right) = 196,451.59$$

- On a calculator,  $PMT = 20,000$ ,  $I/YR = 9$ ,  $N = 25$ ,  $FV = 0$

## Practice Problem

- The FV of the 50 savings, each of the amount  $\$S$ , should be that

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$$FV_{\text{savings, year 50}} = S \left( \frac{(1 + 0.09)^{50} - 1}{0.09} \right) = 196,451.59 \Rightarrow S = \$241.02$$

- On a calculator,  $FV = 196,451.59$ ,  $I/YR = 9$ ,  $N = 50$ ,  $PV = 0$

## Practice Problem

- b. Now the cash flow also carries the \$50,000 college payment in year 20, and,  $PV \text{ of all savings, discounted to today} = PV \text{ of all withdrawals, discounted to today}$

## Practice Problem

- b. Now the cash flow also carries the \$50,000 college payment in year 20, and, PV of all savings, discounted to today = PV of all withdrawals, discounted to today
- The first step is to determine the PV of the cash needed for both college and retirement at time 0

$$PV = \frac{50,000}{1.09^{20}} + \frac{196,451.59}{1.09^{50}} = 11,563.53$$

## Practice Problem

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- Now determine the annual savings for 50 years that equates to that present value

$$\$11,563.53 = C \times \frac{1 - \frac{1}{(1+9\%)^{50}}}{9\%} \Rightarrow C = \$1,054.90$$

- On a calculator, PV = 11,563.53, I/YR = 9, N = 50, FV = 0

## Practice Problem

- Professor's Annuity Corporation offers a lifetime annuity to retiring professors. For a payment of \$79,000 at age 65, the firm will pay the retiring professor \$575 a month until his death
- a. If the professor's remaining life expectancy is 20 years, what is the monthly interest rate on this annuity?
- b. If the monthly interest rate is 1.00%, what monthly annuity payment can the firm offer to the retiring professor?

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$$C \times \left[ \frac{1}{0.01} - \frac{1}{0.01 \times (1.01)^{240}} \right] = \$79,000 \Rightarrow C = \text{PMT} = \$869.86$$

- On a calculator,  $N = 240$ ,  $\text{PV} = -79,000$ ,  $\text{FV} = 0$ ,  $\text{I/YR} = 1$ . Find PMT

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- These have handy formulas for calculating their present values!
- Timing,
  - Ordinary perpetuities and annuities make their first payment at time  $t = 1$
  - Perpetuity and annuity dues make their first payment at  $t = 0$
  - Delayed perpetuities and annuities make their first payment at  $t > 1$

## Module 4, Part 4

# Inflation

# History of Inflation



Note: Shaded area represents recession, as determined by the National Bureau of Economic Research.  
Source: U.S. Bureau of Labor Statistics.

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- **Real interest or real return**, the rate at which purchasing power increases
  - Compare the ability to buy a basket of goods at time  $t = 0$  to the ability to buy the same basket of goods at time  $t = 1$

# Inflation and Returns

- In the presence of inflation, an investor's real return (e.g., the real interest rate) is always less than the nominal return

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- Inflation also affects discounting,
  - We should discount nominal dollar cash flows using a nominal discount rate
  - We should discount real dollar cash flows using a real discount rate

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- Approximation,  $\text{real} \approx 6\% - 2\% = 4.0\%$

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- Perpetuities and annuities have shortcut PV formulas, mind the timing of the first payment
- The real return strips inflation out of the nominal return

# Thank you!

Please let me know if you have any questions.